lecture13 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a program that asks the user to enter a positive integer `n`, and then prints the sum of `1 + 2 + 3 + ... + n` in three different ways:

1. Using the formula: n(n+1)/2. Put this in a function called `sum_formula(n)` that returns (not prints!) the result.

2. Using a for loop. Put this in a function called `sum_loop(n)` that returns (not prints!) the result.

3. Using this recursive formula: if `n` is `1`, then return `1`. Otherwise, return `n + sum_recursive(n-1)`. Put this in a function called `sum_recursive(n)` that returns (not prints!) the result.

```
What is n? 5

  sum_formula(5) returns 15
     sum_loop(5) returns 15
sum_recursive(5) returns 15
```

All the returned numbers should be the same.

Note that the recursive one might fail if `n` is too large, e.g. try running it with `n = 1000`.

Sample Solution

```
def sum_formula(n):
    return n * (n + 1) // 2

def sum_loop(n):
    total = 0
    for i in range(1, n + 1):
        total += i
    return total

def sum_recursive(n):
    if n == 1:
        return 1
    else:
        return n + sum_recursive(n - 1)

# test code
n = int(input('What is n? '))
print()
print(f'  sum_formula({n}) returns {sum_formula(n)}')
print(f'     sum_loop({n}) returns {sum_loop(n)}')
print(f'sum_recursive({n}) returns {sum_recursive(n)}')
```

Question 2 (q2_sol.py)

Write a program that reads a positive integer n from the user, and then prints all the prime numbers less than n, followed by the total number of primes found.

As part of your program, define and use a function called `is_prime(n)` that returns `True` if `n` is prime, and `False` otherwise.

A number is prime if it is in an integer greater than 1 that has no positive divisors other than 1 and itself. The first few primes are 2, 3, 5, 7, 11, ...

```
Enter a positive integer: 100

The prime numbers less than 100 are:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97

Total number of primes less than 100: 25
```

Sample Solution

```
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True

n = int(input('Enter a positive integer: '))
print(f'\nThe prime numbers less than {n} are: ')
prime_count = 0
for i in range(2, n):
    if is_prime(i):
        print(i, end=' ')
        prime_count += 1
print()
print()
print(f'Total number of primes less than {n}: {prime_count}')
```